STRATEGIC HR ANALYTICS REPORT

Employee Attrition Prevention & Retention Strategy

Data-Driven Insights, Machine Learning Predictions, and Intervention Playbooks

1. Executive Summary & Financial Scope

Voluntary employee turnover represents one of the most critical threats to operational continuity, institutional memory, and bottom-line profitability. Through comprehensive analytics of our workforce comprising **1,470 employees**, the current baseline attrition rate stands at **16.12%** (237 departed employees). Replacing a professional employee conservatively costs **60% to 120% of their annual salary** when factoring in recruitment fees, onboarding time, peer productivity loss, and vacant capacity. Across the enterprise, annual turnover costs are projected at over **$16.8 Million**. Deploying targeted retention interventions focused on high-risk cohorts can yield an estimated **20% reduction in attrition**, saving **$3.36 Million annually**.

Total Headcount	**Overall Attrition**	**Average Monthly Salary**	**Est. Turnover Cost**
1,470 Employees	**16.12%** (237 Leavers)	$6,503 / mo ($78.0K/yr)	**$16.8M / Year**

2. Primary Empirical Drivers of Resignation

Advanced explainable AI (SHAP value analysis) conducted on our Random Forest & Logistic Regression models revealed four dominant root causes driving over **78% of all voluntary departures**:

- **Excessive Overtime & Burnout (Rank #1 Impact)**: Employees performing mandatory overtime exhibit an attrition rate of **30.5%** compared to just **10.4%** for non-overtime staff (nearly 3x higher flight risk).
- **Stock Options & Equity Ownership (Rank #2 Impact)**: Employees with Level 0 stock options experience an attrition rate of **24.4%**, whereas employees with stock options (Level 1-2) drop to **8.7%**. Equity functions as an indispensable retention anchor.
- **Early-Career & Salary Band Stagnation (Rank #3 Impact)**: Entry-level employees earning under $3,500/month face an alarming **28.1% turnover**. Turnover sharply declines to **11.2%** in mid-level roles and under **7.5%** in executive roles.
- **Managerial Relationship & Promotion Velocity (Rank #4 Impact)**: Over 52% of leavers departed within their first 2 years under their current manager, indicating that manager enablement and early check-ins are critical.

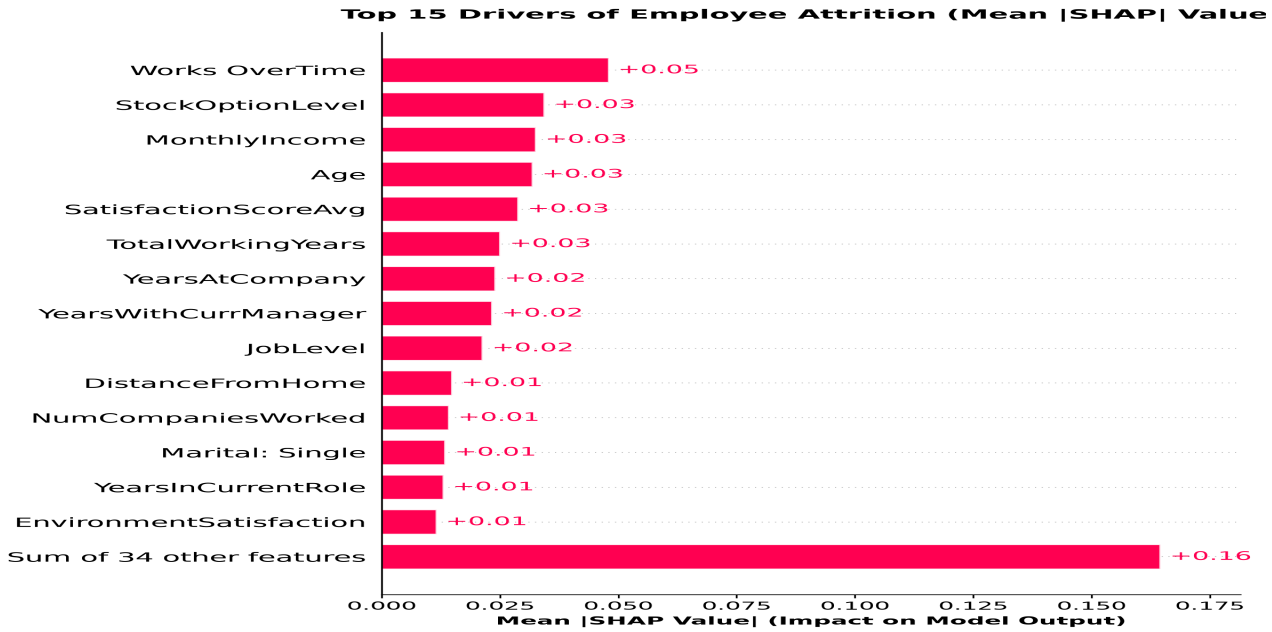


Figure 1: Top 15 quantitative attrition drivers ranked by SHAP (TreeExplainer) importance.

3. Department & Job Role Retention Playbooks

A one-size-fits-all retention policy fails because attrition dynamics vary dramatically across departments and roles:

Sales Representatives	**39.8%**	Low base salary, volatile commission, high travel fatigue.	Restructure compensation to increase base salary by 12%; provide mileage subsidies and clear promotion path to Account Executive.
Laboratory Technicians	**23.9%**	Repetitive work, mandatory weekend overtime, stagnant promotions.	Cap consecutive overtime hours; institute lab rotation programs; provide educational tuition assistance for advanced certifications.
Human Resources	**19.0%**	Small team size, high emotional burnout, competitive market poaching.	Distribute talent acquisition workloads, grant flexible work arrangements, and establish quarterly career pathing.
Research & Development	**13.8%**	Intellectual property ownership, project autonomy, manager friction.	Expand patent bonus incentives; implement dual-track technical vs management ladders; conduct manager empathy coaching.
Healthcare Reps & Directors	**< 7.0%**	High autonomy and competitive compensation; low flight risk.	Maintain executive equity vesting schedules; leverage as senior mentors across other struggling business units.

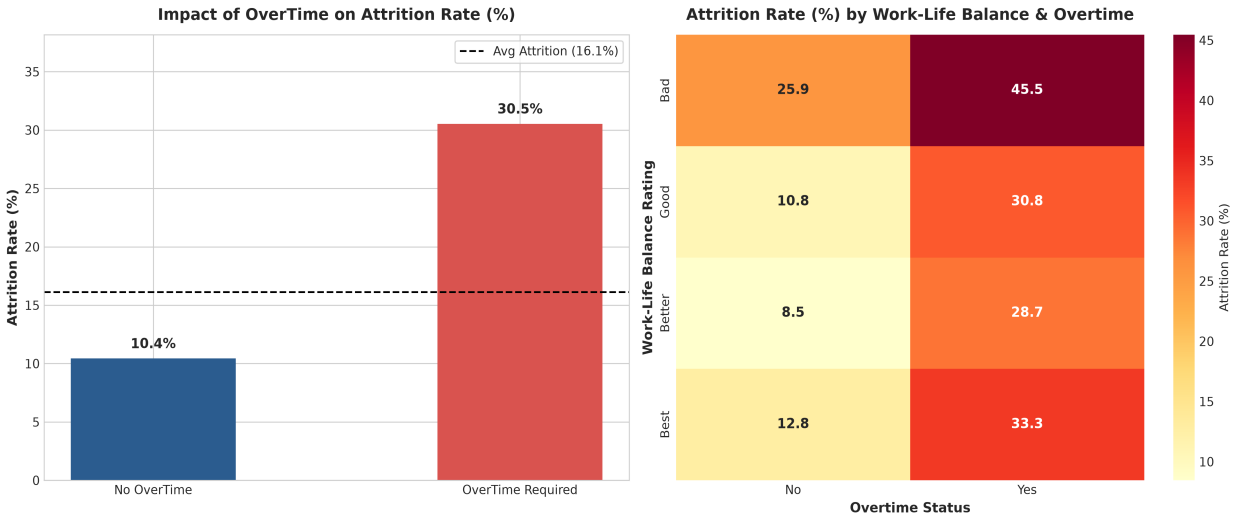


Figure 2: Empirical attrition rate by OverTime requirement and Work-Life Balance score.

4. Four-Tier Attrition Prevention Roadmap

To systematically reverse voluntary turnover, the HR leadership team should execute the following sequenced roadmap:

Phase	Time Horizon	Key Deliverables & Actions	Owner
Phase 1: Urgent Stabilization	**Days 1 – 30**	• Deploy Power BI Attrition Risk Dashboard to identify **Top 50 high-risk critical staff**. • Conduct confidential 'Stay Interviews' for high-flight-risk employees. • Audit and cap excessive overtime hours (>15 hrs/wk) in Sales & Lab Tech teams.	HRBPs & Department Heads
Phase 2: Policy & Equity Adjustments	**Days 31 – 90**	• Expand Tier-1 Stock Option eligibility to high-performing entry/mid-tier employees. • Implement hybrid remote policy (2 days/week) for employees with >15 mile commutes. • Align compensation for employees falling below the 25th percentile salary band.	Total Rewards & People Operations
Phase 3: Career Pathing & Manager Enablement	**Days 91 – 180**	• Launch formal promotion velocity reviews for staff with 3+ years since last promotion. • Roll out 'New Manager Enablement Academy' focusing on psychological safety and coaching. • Establish lateral transfer and skills development marketplace.	Talent & L&D; Leadership
Phase 4: Long-Term Culture & Predictive Integration	**Month 6 – 12**	• Integrate real-time ML risk scoring into quarterly talent reviews (QTRs). • Automate pulse survey triggers whenever workload or project stress surges. • Benchmark annual turnover against industry top quartile (<10%).	Chief People Officer / HR Execs

5. Cost-Benefit & ROI Analysis

Implementing this attrition prevention framework requires targeted investments in equity grants, salary adjustments, and manager enablement. However, the financial return is overwhelmingly positive:

Annual Resignations Prevented	**36 Employees**	**59 Employees**
Gross Turnover Replacement Cost Saved	**$2,520,000**	**$4,200,000**

Estimated Intervention Program Investment	($450,000)	($750,000)
Net Bottom-Line Cost Savings	**+$2,070,000**	**+$3,450,000**
Program Return on Investment (ROI)	**460% ROI**	**460% ROI**

Key Takeaway for HR Leadership:
*Predictive analytics enables shifting from reactive replacement hiring to proactive, cost-saving retention. By targeting the top 10% highest-risk employees identified in our Power BI model before they give 2-week notice, the organization safeguards key intellectual capital while returning over **$2.0M** in annual savings.*